GROUND REACTION FORCE DURING VERTICAL COUNTERMOVEMENT JUMP

Student ID: 1268780
ENGG*3150 F25
School of Engineering, University of Guelph, Guelph, Canada

INTRODUCTION

Bone stress injuries (stress fractures) are very common in sports involving repetitive jumping and landing. These injuries happen from microdamage dealt by high impact loading the pelvic limbs during take-off and landing. A lot of biomechanical studies only focus on the peak ground reaction (GRF) or jump height. This does not consider how loading patterns contribute to bone stress. Cameron et al. (2022) found that poor landing mechanics, measured by the Landing Error Scoring System (LESS), increased the risk of pelvic limb stress fractures by about 15% for each addition landing error [1].

The objective of this study was to analyze and compare the ground reaction force (GRF) and the impulse on take-off and landing (Countermovement jumps - CMJs) in healthy participants. Before conducting the experiment, it was hypothesized that the take-off would be asymmetrical. This is because in most people their dominant leg may produce a greater vertical force and impulse. This shows the natural tendency for most people to rely a lot more on their dominant limbs during explosive movements.

METHODS

The purpose of the laboratory was to study the differences between the GRFs in the right and left legs while undergoing CMJs. Two healthy volunteers (Participant 1; male, age 20, 61.2kg, 1.6m in height) (Participant 5; female, age 20, 71.5kg, 1.65m in height) participated in this study. Both participants had no lower-limb injuries within the past six months and also gave written informed consent before partaking in the experiment.

This lab consists of a coordinate system which was manually defined to keep the anterior-posterior direction towards the x-axis, the medial-lateral direction in the y-axis, and the superior-inferior direction in the z-axis. These axes were manually calibrated before the data collection to make sure all the measured forces correspond to the correct directions. The GRFs collected data using two separate plates: Kistler 9281E (Kistler, Winterthur, Switzerland) (Piezoelectric sensor) and an AMTI AccuGait (AMTI, Watertown, USA) (Electromechanical sensor). These plates were connected to Vicon Data Acquisition (DAQ) software (Vicon Motion Systems Ltd., Oxford, UK) to help synchronize the data being recorded.

Each participant wore comfortable/casual clothing and were given the chance to wear either socks, shoes, or be barefoot for the experiment (condition recorded in the participant information). Before conducting the test, each participant's age, sex, height, and mass were recorded. Both participants were told to complete a “clean” maximal effort CMJ with both their hands placed on their hip to eliminate arm swinging and isolate the lower limbs only.

Both participants (P1 and P5) completed 6 successful “clean” CMJs, three jumps under 2000 Hz and 3 jumps under 30 Hz. The force data was processed in MATLAB R2024a (MathWorks, Natick, MA) and average 3 trials per participant with their respective Hz. Using this software, a table of values for the peak vertical force and impulse during take-off and landing were computed. A fast fourier transform graph (FFT) was also made using this software to evaluate whether the sampling rate chosen for measuring GRF during the CMJ take off and landing was reasonable. The main variables to look for are the peak take-off and landing forces, impulses, and centre of pressure (COP) positions, which are all compared between the participants and between the dominant and non-dominant legs.

RESULTS

Both participants showed clear vertical GRF peaks during both take-off and landing (Fig. 2). At 2000 Hz, P1 had an equal takeoff peak (~1000N/kg) on the dominant leg compared to the non-dominant (~1000N/kg). P5 showed a higher takeoff peak from her left foot (~1200N/kg) compared to her right (~1100N/kg). Both participants showed that their landing peaks were much higher than the take-off

peaks, meaning there was a bigger impact loading while descending. The FFT results show that the 2000 Hz sampling rate accurately captured a lot of detail. The 30 Hz rate lost a lot of high frequency detail which resulted in much easier reading waveforms.

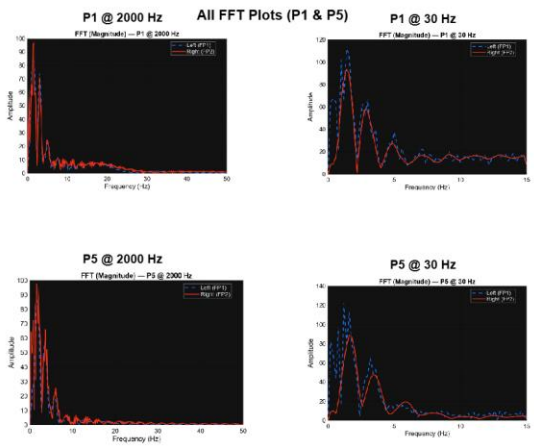

Figure 1: FFT plots comparing vertical ground reaction forces from Kistler (dashed blue) and AMTI (solid red) plates for P1 and P5 at 2000 Hz (left) and 30 Hz (right).

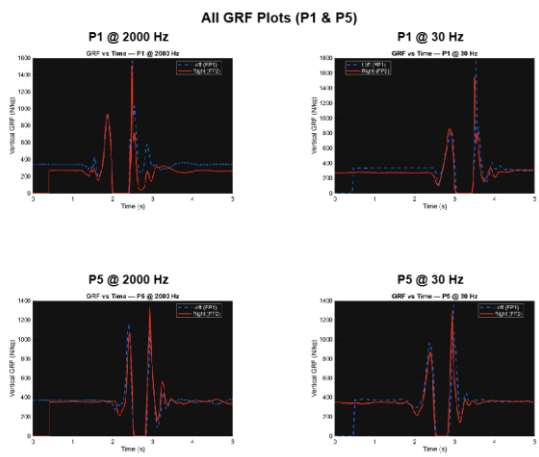

Figure 2: Normalized Vertical GRF vs. time plots of Kistler (dashed blue) and AMTI (solid red) plates for P1 and P5 at 2000 Hz (left) and 30 Hz (right).

DISCUSSION

The vertical GRF showed slight asymmetry between the left and right feet during the take-off and landing. This aligns with the original hypothesis where the dominant leg would be stronger and produce a higher force. This observation supports the scholarly article by Pérez-Castilla A where it shows consistent limb differences in CMJ performance due to habitual force production asymmetry [2]. The landing forces were much higher than the take-off forces in both participants, meaning there was a bigger loading during impact. Reducing the sampling rate only caused the impact peaks to appear lower and easier to read because of the loss in detail. COP positions at take-off and landing were quite similar, meaning there was consistent foot placement and controlled landings across the trials. Potential limitations can include jumping/landing techniques (cannonball/toe-first landing), calibration consistencies between the Kistler and AMTI plates, and zeroing drift or synchronization issues.

CONCLUSIONS

The results supported the hypothesis, showing a slight force of asymmetry between both legs and the dominant limb producing higher GRFs during take-off and landing. In the future, this experiment can include using a higher sampling frequency to improve measurement precision.

REFERENCES

1. Cameron KL, Peck KY, Davi SM, et al. Association Between Landing Error Scoring System (LESS) Items and the Incidence Rate of Lower Extremity Stress Fracture. *Orthopaedic Journal of Sports Medicine*. 2022;10(6).

2. Pérez-Castilla A, et al. Between-session reliability of performance and asymmetry variables obtained during unilateral and bilateral countermovement jumps in basketball players. PLOS ONE. 2021;16(7):e0255458.

Table 1: Average peak take-off and landing forces (N/kg), center of pressure (mm), and corresponding take-off and landing impulses for the left and right legs.

Participant	Peak take-off force (N/kg)	Peak landing force (N/kg)	Impulse take-off	Impulse landing	Center of pressure take-off (x,y)(mm)	Center of pressure landing (x,y)(mm)

	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right	Left	Right
Participant 1	13.4	13.7	22.5	20.4	1.55	1.48	1.26	1.24	299.1,150.9	372.4, -292.3	300.0, 200.0	300.0, -385.0
Participant 5	15.0	13.9	15.3	16.4	1.29	1.02	0.88	1.08	300.0, 200.0	280.1, -321.0	300.0, 200.0	310.3, -329.8